



PRESIDENCY UNIVERSITY

CSE208: Digital Logic Design Laboratory

Class 02: Universal Gates

NAND & NOR Implementation of Logic Gates and Boolean Functions

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Class 02: Learning Objectives

By the end of today's laboratory, students should be able to:

- 1 Explain why NAND and NOR are called **universal gates**.
- 2 Implement a NOT gate using only NAND gates.
- 3 Implement AND and OR gates using only NAND gates.
- 4 Implement a NOT gate using only NOR gates.
- 5 Implement OR and AND gates using only NOR gates.
- 6 Construct Boolean functions using only NAND gates.
- 7 Construct Boolean functions using only NOR gates.
- 8 Verify the designed circuits experimentally.

From Last Class to Today's Class

Last Class

AND, OR, NOT, NAND, NOR, XOR, XNOR



Today's Question

Can we build the other gates using **only one type of gate**?

Answer

Yes.

NAND and NOR are Universal Gates

What is a Universal Gate?

Definition

A **universal gate** is a logic gate from which we can construct the basic gates and, consequently, any Boolean function.

There are two fundamental universal gates:

NAND

NOR

Key Idea

If a gate can be used to construct **NOT**, **AND**, and **OR**, then it can be used to implement arbitrary Boolean functions.

Why Are NAND and NOR Universal?

Any Boolean function can be expressed using:

AND, OR, NOT

Therefore, if a single gate can implement all three:

$NAND \Rightarrow AND, OR, NOT$

and

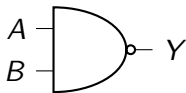
$NOR \Rightarrow AND, OR, NOT$

then:

$NAND$ and NOR are universal

One gate family $\rightarrow$ Complete digital logic

NAND Gate Review



$$Y = \overline{AB}$$

<i>A</i>	<i>B</i>	<i>Y</i>
0	0	1
0	1	1
1	0	1
1	1	0

NAND = NOT + AND

NAND as NOT Gate

Connect both NAND inputs to the same signal:

$$A \text{ NAND } A$$

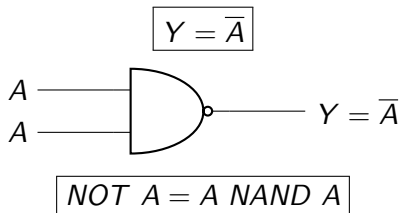
Therefore:

$$Y = \overline{AA}$$

Since:

$$AA = A$$

we get:



NAND as AND Gate

We know:

$$A \text{ NAND } B = \overline{AB}$$

This gives the complement of AND.

So NAND the result with itself:

$$Y = \overline{\overline{AB}} \cdot \overline{\overline{AB}}$$

Since:

$$X \cdot X = X$$

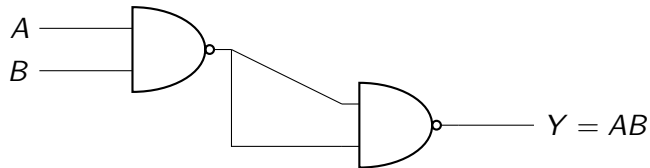
we obtain:

$$Y = AB$$

Therefore:

$$AND = (A \text{ NAND } B) \text{ NAND } (A \text{ NAND } B)$$

NAND as AND Gate — Circuit



$$Y = (A \text{ NAND } B) \text{ NAND } (A \text{ NAND } B)$$

Two NAND gates implement one AND gate.

NAND as OR Gate

We want:

$$Y = A + B$$

Using De Morgan's theorem:

$$A + B = \overline{\overline{A} \cdot \overline{B}}$$

Now generate the complements using NAND:

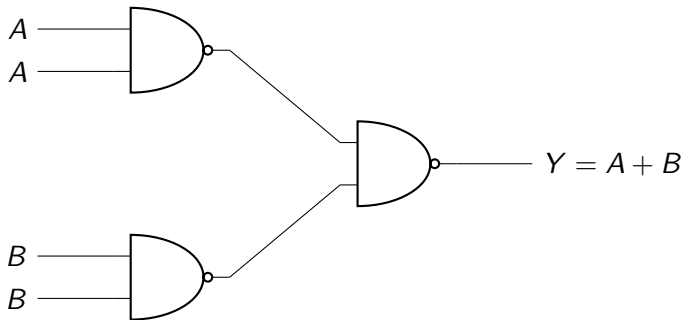
$$\overline{A} = A \text{ NAND } A$$

$$\overline{B} = B \text{ NAND } B$$

Therefore:

$$A + B = (A \text{ NAND } A) \text{ NAND } (B \text{ NAND } B)$$

NAND as OR Gate — Circuit



$$Y = (A \text{ NAND } A) \text{ NAND } (B \text{ NAND } B)$$

NAND-Only Implementation Summary

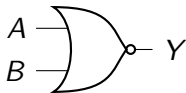
Required Gate	NAND-only Implementation
NOT	$A \text{ NAND } A$
AND	$(A \text{ NAND } B) \text{ NAND } (A \text{ NAND } B)$
OR	$(A \text{ NAND } A) \text{ NAND } (B \text{ NAND } B)$

Conclusion

Because NAND can construct NOT, AND and OR:

NAND is a Universal Gate

NOR Gate Review



$$Y = \overline{A + B}$$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

NOR = NOT + OR

NOR as NOT Gate

Connect both inputs together:

$$A \text{ NOR } A$$

Therefore:

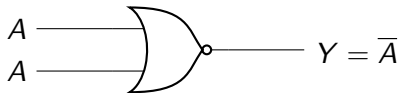
$$Y = \overline{A + A}$$

Since:

$$A + A = A$$

we obtain:

$$Y = \overline{A}$$



$$\text{NOT } A = A \text{ NOR } A$$

NOR as OR Gate

A NOR gate gives:

$$A \text{ NOR } B = \overline{A + B}$$

To obtain OR, complement the result again:

$$Y = \overline{\overline{A + B}}$$

Therefore:

$$Y = A + B$$

Using another NOR gate:

$$OR = (A \text{ NOR } B) \text{ NOR } (A \text{ NOR } B)$$

NOR as AND Gate

Using De Morgan's theorem:

$$AB = \overline{\overline{A} + \overline{B}}$$

Generate the complements:

$$\overline{A} = A \text{ NOR } A$$

$$\overline{B} = B \text{ NOR } B$$

Therefore:

$$AB = (A \text{ NOR } A) \text{ NOR } (B \text{ NOR } B)$$

De Morgan's theorem makes the conversion possible.

NOR-Only Implementation Summary

Required Gate	NOR-only Implementation
NOT	$A \text{ NOR } A$
OR	$(A \text{ NOR } B) \text{ NOR } (A \text{ NOR } B)$
AND	$(A \text{ NOR } A) \text{ NOR } (B \text{ NOR } B)$

Conclusion

Because NOR can construct NOT, OR and AND:

NOR is a Universal Gate

NAND vs NOR

Feature	NAND	NOR
Universal	Yes	Yes
NOT	$A \text{ NAND } A$	$A \text{ NOR } A$
AND	$(A \text{ NAND } B) \text{ NAND result}$	$(A \text{ NOR } A) \text{ NOR } (B \text{ NOR } B)$
OR	$(A \text{ NAND } A) \text{ NAND } (B \text{ NAND } B)$	$(A \text{ NOR } B) \text{ NOR result}$

NAND and NOR can each implement complete logic systems.

From Gates to Boolean Functions

Suppose we are given:

$$F = (A + B)C$$

We can build this using:

$$A + B$$

followed by:

$$(A + B)C$$

Normally:

$$OR \rightarrow AND$$

But today we want:

NAND-Only Boolean Function — Step 1

Given:

$$F = (A + B)C$$

We want to implement this function using **NAND gates only**.

First, generate the complements of A and B :

$$A' = A \text{ NAND } A = \overline{A}$$

$$B' = B \text{ NAND } B = \overline{B}$$

Now NAND these two signals:

$$X = A' \text{ NAND } B'$$

Therefore:

$$X = \overline{A'B'}$$

Using De Morgan's theorem:

$$X = A + B$$

NAND-Only Boolean Function — Step 2

Now combine X with C using a NAND gate:

$$Y = X \text{ NAND } C$$

Therefore:

$$Y = \overline{XC}$$

Substituting $X = A + B$:

$$Y = \overline{(A + B)C}$$

Finally, use one more NAND gate as an inverter:

$$F = Y \text{ NAND } Y$$

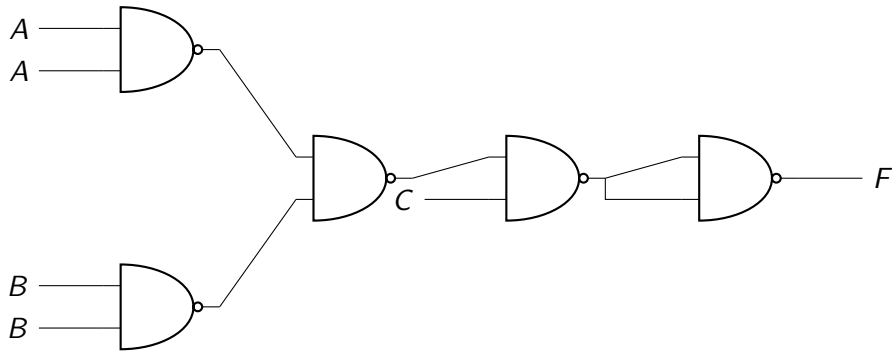
Thus:

$$F = \overline{Y}$$

and therefore:

$$F = (A + B)C$$

NAND-Only Circuit



$$F = (A + B)C$$

NOR-Only Boolean Function — Step 1

Given:

$$F = (A + B)C$$

We want to implement this function using **NOR gates only**.

Using De Morgan's theorem:

$$F = \overline{\overline{(A + B)} + \overline{C}}$$

First, generate:

$$X = A \text{ NOR } B$$

Therefore:

$$X = \overline{A + B}$$

Also, generate the complement of C :

$$Y = C \text{ NOR } C$$

$$Y = \overline{C}$$

NOR-Only Boolean Function — Step 2

Now apply a NOR gate to X and Y :

$$F = X \text{ NOR } Y$$

Therefore:

$$F = \overline{X + Y}$$

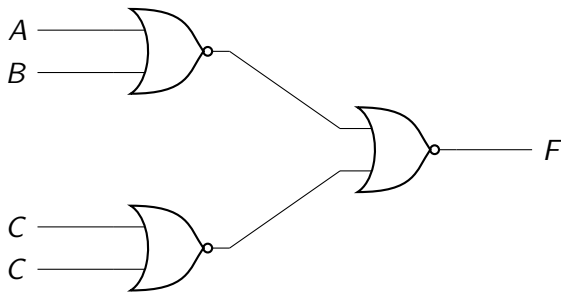
Substituting X and Y :

$$F = \overline{\overline{A + B} + \overline{C}}$$

Using De Morgan's theorem:

$$F = (A + B)C$$

NOR-Only Circuit



$$F = (A + B)C$$

NAND Logic

NAND $\rightarrow$ *NOT*, *AND*, *OR*

NOR Logic

Experiment 2

Realization of Basic Logic Gates Using Universal Gates

Objective

Construct NOT, AND and OR gates using:

NAND only

and

NOR only

Then verify their truth tables.

Part A — NAND Implementation

Students will construct:

- 1 NOT using NAND
- 2 AND using NAND
- 3 OR using NAND

For every circuit:

Construct $\rightarrow$ Apply Inputs $\rightarrow$ Observe Output $\rightarrow$ Verify Truth Table

Record

For each circuit, record:

- Circuit diagram
- Input combinations Observed output Theoretical output
- Verification result

Part B — NOR Implementation

Students will construct:

- 1 NOT using NOR
- 2 OR using NOR
- 3 AND using NOR

For each circuit:

Theory $\rightarrow$ Circuit $\rightarrow$ Test $\rightarrow$ Verification

Challenge

Try to explain the circuit using Boolean algebra, not just by observing the LED.

Boolean Function Implementation

Now implement:

$$F = (A + B)C$$

using:

- ① NAND gates only
- ② NOR gates only

Students should:

- ① Derive the Boolean transformation.
- ② Draw the circuit.
- ③ Construct the circuit.
- ④ Test all input combinations.
- ⑤ Compare experimental and theoretical outputs.

Truth Table Verification

For:

$$F = (A + B)C$$

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

Quick Quiz #1

Which gate is universal?

- ① AND
- ② OR
- ③ NAND
- ④ XOR

Quick Quiz #1

Which gate is universal?

- ① AND
- ② OR
- ③ NAND
- ④ XOR

NAND

Quick Quiz #2

How can you construct a NOT gate using NAND?

$$Y = A \text{ NAND } A$$

Why?

$$Y = \overline{AA}$$

$$Y = \overline{A}$$

Quick Quiz #3

How can you construct an AND gate using NAND?

$$X = A \text{ NAND } B$$

Then:

$$Y = X \text{ NAND } X$$

Therefore:

$$\boxed{Y = AB}$$

Quick Quiz #4

Complete the expression:

$A + B =$

Quick Quiz #4

Complete the expression:

$$A + B = \boxed{\phantom{A \text{ NAND } A \text{ NAND } (B \text{ NAND } B)}}$$

$$\boxed{A + B = (A \text{ NAND } A) \text{ NAND } (B \text{ NAND } B)}$$

Design Challenge

Implement:

$$F = A + BC$$

using:

- ① NAND gates only
- ② NOR gates only

Do not start by drawing gates.

First transform the Boolean expression.

Troubleshooting Checklist

If the circuit does not work:

- ① Check the gate type.
- ② Check the input connections.
- ③ Check whether inputs are floating.
- ④ Check VCC and GND.
- ⑤ Check the IC pin connections.
- ⑥ Check whether the LED/output is connected correctly.
- ⑦ Verify the Boolean expression.
- ⑧ Test one intermediate output at a time.

Best Practice

Debug from Input → Intermediate Nodes → Output

Common Mistakes

Logic Mistakes

- Confusing NAND with AND
- Forgetting the output inversion
- Incorrect De Morgan transformation
- Connecting the wrong intermediate node

Hardware Mistakes

- Wrong IC pin
- Missing VCC/GND
- Floating inputs
- Incorrect LED connection

Understand the logic before troubleshooting the hardware.

What You Should Be Able to Do Now

At the end of today's class, you should be able to:

- 1 Explain universal gates.
- 2 Build NOT using NAND.
- 3 Build AND using NAND.
- 4 Build OR using NAND.
- 5 Build NOT using NOR.
- 6 Build OR using NOR.
- 7 Build AND using NOR.
- 8 Implement a Boolean expression using only NAND gates.
- 9 Implement a Boolean expression using only NOR gates.

Class 02 — Key Takeaways

$NAND \Rightarrow NOT, AND, OR$

$NOR \Rightarrow NOT, OR, AND$

Therefore:

NAND and NOR are Universal Gates

Lab Submission — Experiment 2

Realization of Basic Logic Gates Using Universal Gates

Your report should contain:

- 1 Experiment title and objective
- 2 Required components
- 3 NAND-only NOT, AND and OR circuits
- 4 NOR-only NOT, OR and AND circuits
- 5 Truth tables
- 6 Boolean derivations
- 7 Boolean-function implementation
- 8 Experimental observations
- 9 Verification
- 10 Conclusion

**Can you build any Boolean function
using only NAND gates?**

Or only NOR gates?

Yes!

That is what “Universal” means.



One Gate Can Build Them All

NAND or NOR

Universal Logic

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